

SYSTEMS AND CONTROL – ASSIGNMENT 2

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We consider a linear time-invariant (LTI) system:

$$\dot{x} = A x + B u$$

$$y = C x$$

Where:

- x = state vector (14×1)
- u = input vector (3×1)
- y = output vector (2×1)

The objective is to analyze the system and design a controller and observer.

1. OPEN LOOP STABILITY

Stability is determined from eigenvalues of matrix A .

- If all eigenvalues have negative real parts = Stable
- If any eigenvalue has positive real part = Unstable
- If eigenvalues lie on imaginary axis = Marginally stable

Conclusion:

The system is unstable since some eigenvalues lie in the right half-plane.

Meaning:

Without control, the system states grow and do not settle.

2. ZERO INPUT RESPONSE

For zero input ($u = 0$):

$$\dot{x} = A x$$

Solution:

$$x(t) = e^{At} x(0)$$

Output:

$$y(t) = C x(t)$$

We plot y_1 vs y_2 (phase portrait).

Observation:

Trajectories move away from origin.

Conclusion:

Confirms instability of the open-loop system.

3. PBH TEST – STABILIZABILITY

Condition:

$$\text{rank} \begin{bmatrix} \lambda I - A & B \end{bmatrix} = n$$

for all eigenvalues λ with $\text{Re}(\lambda) \geq 0$

Meaning:

All unstable modes must be controllable.

Result:

System is stabilizable.

Interpretation:

Even though system is unstable, control input can stabilize it.

4. PBH TEST – DETECTABILITY

Condition:

$$\text{rank} \begin{bmatrix} \lambda I - A \\ C \end{bmatrix} = n$$

for all eigenvalues λ with $\text{Re}(\lambda) \geq 0$

Meaning:

All unstable modes must be observable.

Result:

System is detectable.

Interpretation:

State estimation is possible using outputs.

5. CONTROLLABILITY & OBSERVABILITY

Controllability Matrix:

$$C_o = \begin{bmatrix} B & AB & A^2B & \dots & A^{(n-1)}B \end{bmatrix}$$

Observability Matrix:

$$O_b = \begin{bmatrix} C \\ CA \\ CA^2 \end{bmatrix}$$

...

$$CA^{(n-1)}]$$

- Rank(Co) = number of controllable states
- Rank(Ob) = number of observable states

6. KALMAN DECOMPOSITION

The system is decomposed into four parts:

- Controllable & Observable
- Controllable & Unobservable
- Uncontrollable & Observable
- Uncontrollable & Unobservable

Using transformation:

$$\tilde{A} = T^{-1} A T$$

$$\tilde{B} = T^{-1} B$$

$$\tilde{C} = C T$$

From this, we extract:

$$A_{co}, B_{co}, C_{co}$$

Key Idea:

Only the controllable–observable subsystem is used for controller design.

7. REDUCED ORDER OBSERVER

Observer equation:

$$\dot{\hat{x}} = A \hat{x} + B u + L (y - C \hat{x})$$

Error:

$$e = x - \hat{x}$$

Error dynamics:

$$\dot{e} = (A - L C) e$$

Goal:

Choose L such that (A – LC) is stable.

8. LYAPUNOV BASED OBSERVER DESIGN

Lyapunov equation:

$$A^T W + W A = -Q$$

$$(Q > 0)$$

Observer gain:

$$L = W^{-1} A_{12}^T$$

Purpose:

Ensures estimation error converges to zero.

9. LQR CONTROL

Cost function:

$$J = \int (x^T Q x + u^T R u) dt$$

Where:

- $Q \geq 0$ = penalizes state
- $R > 0$ = penalizes control effort

Optimal control:

$$u = -K x$$

10. ALGEBRAIC RICCATI EQUATION

$$A^T P + P A - P B R^{-1} B^T P + Q = 0$$

Gain:

$$K = R^{-1} B^T P$$

11. HAMILTONIAN METHOD

Hamiltonian matrix:

$$H = \begin{bmatrix} A & -B R^{-1} B^T \\ -Q & -A^T \end{bmatrix}$$

Steps:

1. Compute eigenvalues of H
2. Select eigenvectors with negative real part
3. Partition into V1 and V2
4. Compute:

$$P = V2 \times \text{inv}(V1)$$

12. CLOSED LOOP SYSTEM

$$\dot{x} = (A - B K) x$$

Condition:

If eigenvalues of $(A - BK) < 0 \rightarrow$ system is stable

13. OUTPUT PHASE PORTRAIT

$$y = C x$$

Plot y_1 vs y_2 .

Observation:

Trajectories converge to origin.

Conclusion:

Closed-loop system is stable.

14. OBSERVER + CONTROLLER

If full state is not available:

$$u = -K \hat{x}$$

Combined system:

$$\dot{x} = A x - B K \hat{x}$$

$$\dot{\hat{x}} = A \hat{x} + B u + L (y - C \hat{x})$$

Result:

Simultaneous state estimation and control.

PROBLEM 2 – INVERTED PENDULUM

The inverted pendulum is inherently unstable.

MODEL

State-space form:

$$\dot{x} = A x + B u$$

States represent:

- Arm angle
- Arm velocity

- Pendulum angle
- Pendulum velocity

LQR CONTROL

Same cost function:

$$J = \int (x^T Q x + u^T R u) dt$$

Control:

$$u = -K x$$

Gain K is obtained using Hamiltonian method.

SIMULATION & ANIMATION

The system is simulated using numerical integration.

- Initial condition slightly disturbed from upright position
- Arm and pendulum motion visualized in 3D

Observation:

- Pendulum returns to upright position
- Oscillations decay over time

Conclusion:

LQR successfully stabilizes the inverted pendulum.

FINAL CONCLUSION

- The system is unstable in open loop
- It is stabilizable and detectable
- Kalman decomposition extracts relevant subsystem
- Observer estimates internal states
- LQR provides optimal control
- Closed-loop system becomes stable
- Inverted pendulum is successfully stabilized